



University of St.Gallen

Institute of Computer Science

Assignment 1- Part 1

This deck compiles the assignment tasks published previously with the lecture slides together with submission instructions.



From insight to impact.

Introduction

Assignment 1 Introduction

You are asked to analyze a real-life event log. The event log can be downloaded [here](#).

The event log records the process of managing road traffic fines by a local police in Italy. This process is regulated by Italian law.

A brief description of the process activities and log data attributes follows. This is some, but limited domain knowledge. Feel free to share your own experience or domain knowledge about how traffic fines are handled in Italy.

Otherwise, it is up to our process mining skills to contribute to the **two main goals**:

- understand what precisely has happened according to the log and
- what improvements to the process and to the data collection we can suggest. This includes improvements to what data is collected, in what form it is represented and fixing any data quality issues that you might find.

Process Activity Description

Activity	Description
Create Fine	The initial creation of the fine in the information system. It initializes the event log attributes <i>amount</i> , <i>dismissal</i> , <i>points</i> and <i>totalPaymentAmount</i> .
Send Fine	A notification about the fine is sent to the offender.
Insert Fine Notification	The notification is received by the offender.
Add penalty	An additional penalty is applied.
Payment	A payment made by the offender is registered.
Send for Credit Collection	Unpaid fines are sent for credit collection. A separate process is started by a collection agency to collect the money of the unpaid fines.
Insert Date Appeal to Prefecture	The offender appeals against the fine to the prefecture*.
Send Appeal to Prefecture	The appeal is sent to the prefecture by the local police.
Receive Result Appeal from Prefecture	The local police receives the result of the appeal. **
Notify Result Appeal to Offender	The local police informs the offender of the appeal result. **
Appeal to Judge	The offender appeals against the fine to a judge.

* A prefecture in Italy is an administrative body representing the national government in each province.

** If the prefecture dismisses the fine, the appeal is deemed accepted, and the obligation to pay the fine is cancelled.

Event Log Attributes Description

Attribute	Description
amount	The amount due to be paid for the fine (in case a penalty is added, the value includes the penalty amount as well as the initial fine amount).
article	The number of the article of the Italian road-traffic law that is violated by the offender (e.g., article 157 refers to stopping and parking vehicles).
dismissal	A flag indicating whether and by whom the fine is dismissed. It is initialized to NIL on creation of the fine. We know the meaning of: - G: dismissed by the judge - #: dismissed by the prefecture
expense	The additional amount due to be paid for postal expenses.
lastSent	N/A
matricola	N/A
notificationType	A flag indicating to whom the fine refers. P: car owner, C: car driver.
resource	A numeric code indicating the employee who handled the case in the corresponding activity.
paymentAmount	The amount paid by the offender in one transaction.
points	Penalty points deducted from the driving license. In Italy, each driver starts with 20 points on their license and may lose points for each offence, based on the gravity.
totalPaymentAmount	The cumulative amount paid by the offender.
vehicleClass	A flag indicating the kind of vehicle driven or owned by the offender.

More General Instructions for Assignment 1

Use a Python notebook (template on Canvas) for the assignment as your main report (which you submit after weeks 3 and 6). Besides the code, your textual answers, the generated output such as charts, you can also include screenshots from external tools. Structure the notebook for readability, cf. screenshot.

By default, create Python code to answer any question unless indicated otherwise in the task explicitly.

Before executing any code, form a *hypothesis*, i.e., think about what the answer could be or in what range you expect the answer to be and compare with the answer you are getting from the code. If you find the answer surprising, say so in your report. You might also want to double check the correctness of your code. For some tasks, we ask you to write down a hypothesis explicitly.

If the answer raises a new question, also write the question down, even if you might not have time to investigate it. If the answer implies some suggestion for improving the process or the data collection (or the data representation), write it down as a finding. Sometimes, we ask explicitly for such a reflection, but even if we don't, it will help to solve some tasks later if you do.

```
1 import marimo as mo
2 import pandas as pd
3 import pm4py

event_log_from_disk = pm4py.read_xes('event_log_from_disk')
```

amount	totalPaymentAmount	time:time
float64 (nulls: 331240) min: 0 max: 8,000 unique: 206	float64 (nulls: 333499) min: 0 max: 4,021 unique: 1,009	datetime64[ns] min: Jan 1, 2000 max: Jun 1, 2012 unique: 1,009
62.59		0 1999-12-
62.59		0 2000-01-
31.30		0 2000-01-
31.30		0 2000-01-
31.30		0 2000-01-

561,470 rows, 16 columns

Session 1

Task 1.1

- a) ...
- b) ...
- c) ...
- d) ...

Session 2

Task 2.1

Session 1

Assignment 1: Tasks from Session 1 (First Event Log Exploration)

1. Load the event log using pm4py (see assignment template on canvas). Answer the following questions [Code optional]:
 - a) What time interval does the log cover?
 - b) How many distinct values for the attribute *vehicleClass* does the log have?
 - c) What are *min*, *median*, *max* of the *initial value* of the attribute *amount* (associated with a *Create Fine* event) ?
Reflect the result: what do you recommend to be investigated further?
 - d) How many events have a value > 0 for the attribute *points*? What are the associated activities? How many case are affected?

Session 2

Assignment 1: Notice

Keep in mind that you are studying a real-world event log, not an academically polished exercise log. This means:

- data might be missing or inconsistent
- events might not always exactly mean what you initially assume
- we do not know all details of the data collection and event log preparation process
- the given description (in Session 1) is what the data providers have published

This of course does not mean that every unexpected observation can be attributed to dirty log data. In fact, most strange observations will make sense in the end.

Assignment 1: Tasks from Session 2, Part 1

(Schema Analysis, Case Inspection)

1. Which granularity do the timestamps of the log have? *[Code welcome but not required.]*
2. Inspect the schemas of all activities. *[Code welcome but not required.]*
 - a) Which activity schemas have attributes with missing values? How many events are affected for each such attribute?
 - b) Which non-global attributes are shared, i.e., local to more than one activity schema? Classify each such attribute that is also numerical as either cumulative or incremental.
3. Select a case of the log with more than three events. Write a short bullet point for each event that summarizes what happened paying attention to the timestamps and all attributes. *[Code welcome but not required.]*

Assignment 1: Tasks from Session 2 Part 2 (*Event enrichment*)

4. Add an attribute to the event log that represents, for each case, the cumulative sum of payment amounts after each event.
 - a) Select and inspect a case (including this new enrichment) to verify that the enrichment was computed correctly.
 - b) Recall that the already existing attribute *totalPaymentAmount* also claims to represent the cumulative sum of payment amounts. Verify that *totalPaymentAmount* was computed correctly by comparing it with the attribute you just added. If you find a deviation, quantify the deviation and inspect and interpret a deviating case.
5. Add an attribute *outstanding_amount* to the event log that represents, for each case, the amount that still has to be paid by the offender after each event. Recall that the offender has to pay the *initial fine amount*, the *penalty* if added and any *expenses* that are recorded. Define *outstanding_amount* and any preliminary attributes as event enrichments.
 - a) Select and inspect a case (including this new enrichment) that the enrichment was computed correctly.
 - b) What is the number of events that have an outstanding amount > 0 ?

Assignment 1: Tasks from Session 2 Part 3 (*Case enrichment*)

5. Create an initial case log that includes all *case attributes*. Add an attribute to the case log that represents the *initial fine amount* (i.e., the *amount* associated with the *Create Fine* event, or equivalently, the first value of *amount* of each case).
 - a) Formulate a hypothesis about the distribution of the initial fine amount.
 - b) Create a suitable visualization of the distribution of the initial fine amount. Describe briefly two interesting observations about the distribution.
6. Create a visualization relating initial fine amount and the *final fine amount* of each case. The *final fine amount* can be defined as the last value of the attribute *amount* for each case. Briefly describe and interpret the relationship that you see.
7. Find at least one noteworthy fact about the distribution (or a relationship) for any of the following attributes: *org:resource*, *dismissal*, *article*, *points*, or *expense*.

Session 3

Assignment 1: Tasks from Session 3 (Outcome Analysis)

1. Define three business level process outcomes for the process under investigation.
 - a) Enrich the case log with Boolean indicators for each outcome and compute the number of cases for each outcome, for each combination of two outcomes as well as the number of cases that have no outcome defined.
 - b) Inspect and interpret a case for each of the three outcomes.
 - c) If existent, inspect and interpret a case that has more than one outcome.
 - d) If existent, inspect and interpret a case that does not belong to any of those outcomes.

Session 4

Assignment 1: Tasks from Session 4 Part 1 (Discovery)

1. Inspect the distribution of sequential variants (in *Disco* or *Python*).
 - a) What is the minimum number of sequential variants that is needed to cover 80% of the cases? *[no code required.]*
 - b) Summarize each of the top 4 sequential variants in one sentence. *[no code required.]*
 - c) The **fine object** sub-log is defined by filtering out all events from the event log except the events from the following activities: ['Create Fine', 'Send Fine', 'Insert Fine Notification', 'Add penalty', 'Send for Credit Collection', 'Payment']. Apply this filter to the event log and compare the number of sequential variants after and before filtering. *[no code required.]*
2. Inspect the process map in *Disco* with absolute frequency or the Advanced Process Map (*Linear Continuous Process Mapper*) **for the fine object defined in task 1** above.
 - a) Write down at least three observations that are unexpected to you and hence you would suggest them to be further investigated. Explain briefly why.
 - b) Use a filter to find a sequential variant that exhibits an unexpected behavior. Report the variant as a sequence of activities.
 - c) Inspect a case of the variant reported in b). Can you come up with an explanation for the unexpected behavior?

Assignment 1: Tasks from Session 4 Part 2 (BPMN)

3. Create a workflow graph model in BPMN that reflects your current (preliminary) understanding of how the ***fine object*** should behave (*You might have seen behavior in the log that does not comply with that understanding. That is ok.*) Use bpmn.io to create a *bpmn* file and a *svg* file, start from the template provided in Canvas, which contains all activities that the fine object is involved in. Make sure that your model contains all these activities and make sure your model is sound.

Submission Instructions

Submission Instructions

- Re-run your notebook to make sure you have no remaining errors.
- Include the following files in a single *zip file* and submit the zip file by the submission deadline on Canvas:
 - the notebook,
 - if you used Marimo, export the notebook to (static) html, to save the outputs of your cells and include the html file in the zip file
 - if you used Jupyter, save your notebook with all cell outputs included.
 - any additional files that your notebook might depend on, e.g.,
 - any helper files with code you might have created or modified,
 - any screenshot files you might have included in your notebook
 - the *bpmn* file used in Task 5.3 and the associated *svg* file
 - Do **not** include the original event log.

Name the zip file `<your_name>.zip` and upload to Canvas.